Supplementary Information

Four-View Visual–LiDAR Loop Closure for Drift Correction in Multi-Storey Indoor Mapping

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Reference numbers refer to the revised main manuscript.

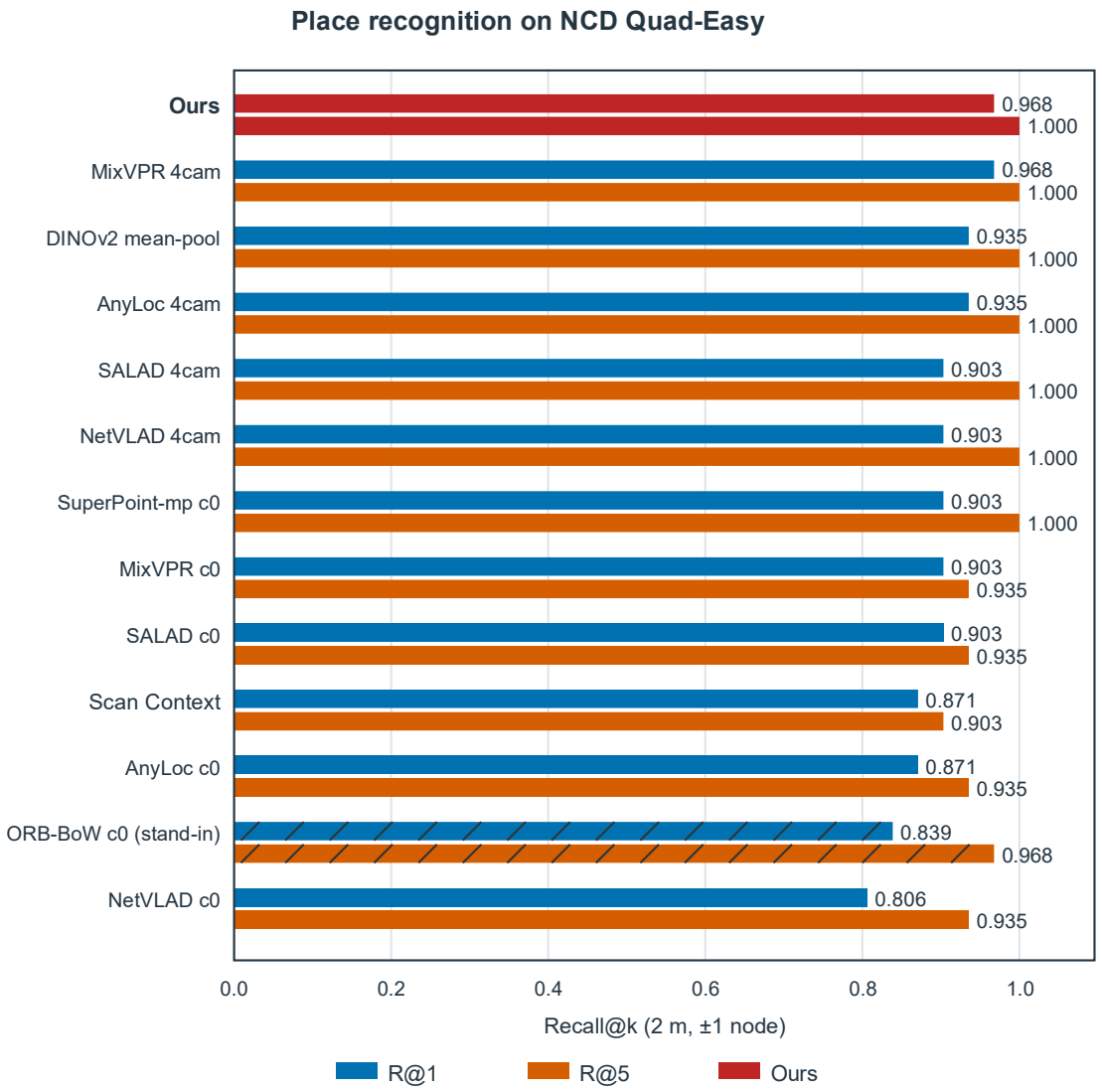
Supplementary Note: Public-dataset retrieval

We evaluated retrieval on the multi-camera Quad-Easy sequence from the Newer College Dataset [41]. The sequence contains four-camera, LiDAR, and IMU observations collected with a handheld rig outdoors over approximately 198 s. We formed 66 nodes at approximately 3 s intervals, used ground-truth poses as the input geometry, and assigned all nodes to one floor.

The evaluation protocol uses a 2 m spatial threshold and a 30 s temporal exclusion, giving 38 ground-truth positive pairs and 31 queries. Every retrieval method in Table 3 is evaluated on these same queries. Under exact-node matching, the retrieved historical node must itself be a ground-truth positive. The tolerance-based rule also accepts either of its immediate neighbours in the historical sequence. Since nodes are approximately 3 s apart, a neighbouring node can lie beyond the 2 m threshold. We report R@1 under both rules and R@5 with the ±1-node tolerance.

The comparison includes LiDAR retrieval with Scan Context and visual retrieval with DINOv2 and MixVPR [16, 27, 28]. The four-camera MixVPR configuration takes the maximum similarity over physical-camera pairs; its c0 configuration uses that camera alone. The DINOv2 baseline mean-pools descriptors from the 12 rectified views. Our retrieval stage uses the joint cyclic scoring described in the main manuscript.

With the ±1-node tolerance, our method and four-camera MixVPR both achieve R@1 of 30/31 and R@5 of 31/31. Scan Context gives the highest exact-node R@1 at 26/31, compared with 17/31 for our method. The difference between the two matching rules reflects whether a method retrieves the reference node or one of its neighbours along the recorded route. Supplementary Figure S1 shows the expanded comparison, with additional implementation details in the Supplementary Methods, “Reproducibility”.



Supplementary Figure S1. Place recognition on NCD Quad-Easy across 13 evaluated configurations. Recall is evaluated on 31 queries under the shared 2 m / 30 s protocol with a tolerance of ±1 keyframe. Within each method, the upper and lower bars show R@1 and R@5, respectively; both bars for Ours are highlighted in red. Ours and four-camera MixVPR both achieve 30/31 R@1 and 31/31 R@5. Hatched ORB-BoW bars denote an OpenCV ORB plus k-means vocabulary stand-in fitted on this sequence. The figure expands the baseline selection summarized in Table 3.

Downstream verification and optimization retain 100 loops. With a tolerance of ±1 keyframe at each endpoint, 96 of the 100 loops match ground-truth positive pairs, covering 32 of the 38 pairs. None of the five opposite-heading positive pairs is covered.

Supplementary Note: Fixed-camera ranking controls

This experiment examines how the camera used for ranking affects the final loop set on the four-storey sequence. Each configuration scores candidates using one physical camera, then applies the same four-camera verification and backend processing. Supplementary Table S1 reports the accepted loops on each floor.

Supplementary Table S1. Fixed-camera ranking.

Ranking	F1	F2	F3	F4	Total
c0 ranking	6	3	5	0	14
c1 ranking	6	0	3	3	12
c2 ranking	9	0	6	0	15
c3 ranking	8	0	5	0	13

Note. Final loops on the four-storey recording.

Ranking with c0, c1, c2, and c3 yields 14, 12, 15, and 13 loops, respectively. Every single-camera ranking configuration leaves at least one floor without a loop. The c1 configuration retains three F4 loops but none on F2; the other configurations retain no F4 loops. In the main experiment, joint four-view ranking retains 26 loops across all four floors. These results show how the ranking input changes which parts of the route receive loop constraints.

Supplementary Methods: Single-storey region-wide alignment

Supplementary Table S2 summarizes alignment between the first and return passes in the corridor and the broader start/finish region. Each metric is the median absolute gap between corresponding surface estimates. The common cells or locations are held fixed between the uncorrected and corrected reconstructions. We also report how many of those locations have a smaller absolute gap after correction.

Supplementary Table S2. Single-storey map alignment.

Region metric	Before (cm)	Corrected (cm)	Common cells or sites	Improved cells or sites
Corridor ground	21.00	0.39	206	183
Corridor upper surfaces	20.76	0.38	62	62
Corridor wall-normal gap	3.73	1.76	62	38
Broad start/finish room ground	34.83	7.32	108	97

Note. Median absolute two-pass gaps in centimetres, evaluated at common cells or sites.

Supplementary Note: Corridor ground extraction

Both reconstructions use the same full-resolution points in the same order. We select outgoing and return observations from the timestamp intervals of nodes 14–20 and 115–120, respectively, adding a 1 s margin at each boundary. The corridor occupies $Y \in [-26.5, -11.0)$ m in the map frame. Its centreline is interpolated from the outgoing trajectory.

For ground extraction, we retain points within 1.7 m laterally of the centreline and within $-1.7 < Z < 0$ m. The corridor is divided into 0.5 m longitudinal and lateral cells, each requiring at least 35 points. Within a cell, points whose heights are within 0.08 m of the 15th-percentile height form the ground subset. This subset must contain at least 15 points, and its median height defines the ground height for that cell.

The signed gap is the return-pass height minus the outgoing-pass height. The summary uses 206 cells available in both passes and both reconstructions. For the profile at each of the 31 longitudinal stations, we first take the median of the cell heights for each pass and then calculate their difference. Figure 3 shifts the longitudinal origin by +26.5 m; the plotted vertical differences are unchanged.

Supplementary Note: Upper-surface and lateral-wall extraction

We use the same pass intervals and corridor centreline as in the Supplementary Note, "Corridor ground extraction". The evaluation region covers $Y \in [-26.5, -11.0)$ m, extends 3 m laterally from the centreline, and retains points within $-2 < Z < 2.7$ m. We divide the corridor into 31 stations at 0.5 m intervals. At each station, the left and right strips occupy lateral coordinates $[-2, -1)$ m and $[1, 2)$ m.

For upper surfaces, points within $1.8 \le Z < 2.6$ m are grouped into 0.02 m height bins. Each strip requires at least 100 points, with at least 30 points in the most populated bin. The upper-surface height is the median of points within 0.04 m of that bin's centre. These observations can include wall–ceiling junctions.

For walls, we select points within $-0.2 \le Z < 1.4$ m and require at least 100 points per strip. The median lateral coordinate represents the wall position. The wall-normal gap is the difference between the return-pass and first-pass median coordinates.

Both metrics are evaluated at station–side locations present in both passes and both reconstructions. This gives 62 common locations for each metric. Supplementary Table S2 reports the median absolute gap and the number of locations whose absolute gap decreases after correction.

Supplementary Note: Broad start/finish ground region

The broader ground comparison uses nodes 0–7 and 124–133, with a 1 s margin at each timestamp boundary. The map-frame region is $X, Y \in [-4, 4)$ m and $-3 < Z < 1$ m. Ground heights are extracted in 0.5 m cells using the percentile-based procedure in the Supplementary Note, "Corridor ground extraction". The summary in Supplementary Table S2 uses the 108 cells present in both passes and both reconstructions.

Supplementary Methods: Implementation parameters

Supplementary Table S3 lists the settings for image rectification, retrieval, pose verification, sequence selection, ground fitting, and graph optimization. Both indoor recordings use a 24 s query-side sequence gap and a 36 s historical-side gap.

Supplementary Table S3. Implementation settings.

Stage	Setting
Rectification	Three 512 × 512 pinhole views per camera, yaw −50°/0°/+50°, 90° FoV
Retrieval	Minimum pair separation 35 s; up to 12 same-floor and 3 unknown-floor candidates; at most 2 per 12 s historical bin
Score normalization	Camera-pair mean/std; floor-specific when at least 32 eligible pairs; std floor 10^{-6}
PnP hypotheses	AP3P-RANSAC per camera, 3 seeds, up to 2000 iterations, confidence 0.999; 0.8° joint angular inlier threshold

Stage	Setting
Visual gate	At least 15 total PnP inliers; at least 2 cameras with at least 5 each
Odometry agreement	Translation ≤5 m; rotation ≤20°
LiDAR geometry	Coverage ≥0.25 and point-to-plane RMSE ≤0.18 m in both directions; refinement ≤0.5 m and ≤5° from PnP
Reverse geometry evaluation	0.2 m voxel grid; 0.3 m nearest-neighbour limit; planarity >0.3
Sequence	At least 3 pairs; minimum endpoint span 6 m on both sides; query gap 24 s, historical gap 36 s; correction agreement ≤0.8 m/8° for every pair of members
Spatial anchors	Redundancy requires both time differences <12 s and both position differences <3 m; prioritize PnP inlier count
Ground plane fitting	Height band −2.6 to −1.2 m along the initial up axis, horizontal radius <12 m; up to 6000 sampled points and 200 RANSAC trials; 0.04 m plane inlier band; ≥150 inliers
Near-horizontal normal residual	When abs(n_z) <0.2 the implementation returns a fixed large residual; the observation remains in the objective
Stable floor segments	Local time window ±9 s; height range ≤0.6 m and slope ≤0.07; ≥5 samples, ≥12 s duration, ≥6 m travel; ground RMSE ≤0.06 m and tilt ≤12°; same-level tolerance 0.9 m, between-level separation ≥1.8 m
Floor-height prior	Initial identified level; standard deviation 0.5 m
Transition measurement	Difference of consecutive stable segment centres; sigma=max(0.3 m, half the sum of segment P90–P10 spreads)
Graph noise	Odometry sigmas 0.01 rad and 0.05 m; ground horizontal-normal sigmas corresponding to 1.5° and height sigma 0.08 m; Huber parameter 1.345
ICP iteration	Up to 30 iterations; correspondence distance decreases from 1.0 to 0.3 m; Cauchy weighting
Pose validity	Finite homogeneous 4 × 4 transform; rotation orthogonality tolerance 10^{-5}; determinant ≥0.99
GNC	Squared whitened-residual cutoff 16.81; continuation multiplier 1.4; up to 60 updates; stop when all weights are within 0.001 of 0 or 1 and μ >50; final loop retained only if weight >0.5, followed by re-optimization

Note. Both indoor recordings use the 24 s query-side gap.

The Hessian-based covariance approximation for point-to-plane registration [40] is transformed into the graph residual frame and regularized before conversion to an information matrix. Additional variance is assigned to weakly constrained translation directions. For n anchors from the same floor and sequence group, each information matrix is scaled by min(1, 3/n) before GNC [20] updates the loop weights. The separate single-storey stress test uses a shared pool of 55 measurements with floor factors disabled, as detailed in the Supplementary Methods, “Reproducibility”.

Supplementary Note: Storey separations

Supplementary Table S4 compares adjacent-floor separations obtained by transforming the same ground observations with each method’s poses. The measured reference separations are 4.201, 4.255, and 4.169 m for F1–F2, F2–F3, and F3–F4. The proposed method gives signed errors of −2.2, +7.9, and +1.4 cm, with a

mean absolute error of 3.83 cm. Using the same 26 loops without floor factors gives a mean absolute error of 14.90 cm. This fixed-loop comparison relates the change in mapped floor separation to the floor constraints.

Supplementary Table S4. Mapped storey separations and measured reference heights (m).

Floor pair	Uncorrected trajectory	Scan Context + ICP	Scan Context + ICP (Cauchy)	Ours	w/o floor factors	Reference height (m)
F1–F2	4.135	3.958	4.012	4.179	4.075	4.201
F2–F3	4.456	4.922	4.767	4.334	4.469	4.255
F3–F4	4.219	4.007	4.224	4.183	4.276	4.169

Note. All methods transform the same ground observations. The separations are computed from ground observations transformed into the map frame. Reference heights are the means of repeated measurements made with a DLX-MD50 laser distance meter at the centres of the ground-observation regions on each floor. Its nominal accuracy is ±(2 mm + D/10000), where D denotes the measured distance in consistent units. Signed errors are mapped separation minus the corresponding reference. Ours gives −2.2, +7.9, and +1.4 cm for F1–F2, F2–F3, and F3–F4, respectively. The mean absolute error across the three floor pairs is 3.83 cm.

Supplementary Methods: Reproducibility

The expanded retrieval comparison includes AnyLoc, SALAD, and NetVLAD [26, 29, 30]. We use AnyLoc with VLAD-DINOv2 ViT-g/14 and an urban vocabulary of 32 clusters, SALAD v1.0.0 with dino_salad.ckpt, MixVPR with a ResNet-50 backbone and 4096-dimensional descriptors, and Pittsburgh-trained VGG16-NetVLAD. For each method, the four-camera configuration takes the maximum cosine similarity over the 4 × 4 physical-camera pairings. The c0 configuration uses that camera alone.

The SuperPoint mean-pool baseline uses at most 512 keypoints per image, extracted with SuperPoint [14]. Their descriptors are averaged, L2-normalized, and compared by cosine similarity. The ORB-BoW configuration uses ORB features [43] extracted with OpenCV and a 1024-cluster k-means vocabulary fitted on the evaluated sequence.

The public code and experiment records are available in release commit 288c020630f790a12518f27490ab115965bfbf11 of the project repository. REPRODUCIBILITY.md describes the input formats and run commands; configs/runtime.yaml and configs/camera.yaml provide the runtime settings and sensor calibration. Both indoor recordings use a 24 s query-side sequence gap and a 36 s historical-side gap, with the two-camera visual-support requirement.

The general entry point fits ground observations from the supplied submaps and passes sequence-supported anchors to graph optimization. The experiment scripts and configurations are provided under research/experiments.

The single-storey sequence/GNC stress test uses a separate input pool of 55 measurements. It requires at least 10 PnP inliers, omits the two-camera participation requirement, and disables floor factors. All four

stress-test configurations share this pool and its covariances. The main single-storey reconstruction uses the full pipeline and contains 17 accepted loops.

Both indoor experiments use PyTorch 2.14.1, OpenCV 5.0.0.93, and GTSAM 4.3.0. Camera–LiDAR extrinsics are calibrated with FAST-Calib [39]. Static image masks combine temporal image variation, the valid fisheye region, manually outlined cable regions, and exclusion of viewing rays above 25° elevation in the gravity-aligned frame.

Each four-storey submap accumulates approximately 3 s of LiDAR data and uses the first valid point timestamp as its pose anchor. The final voxel size is 0.12 m. The frontend uses a 1.0 m LIO voxel and 0.5 m surface filtering, with cross-camera tracking and exposure estimation enabled.

The public repository (https://github.com/yuanbanlin8-dotcom/four-view-visual-lidar-loop-closure) contains indoor-flight candidate pairs, trajectories, loop decisions, ground measurements, comparison and ablation outputs, and NCD evaluation inputs and results. DATA.md describes their locations and access conditions. For indoor ROS bags and dense point clouds, contact the repository maintainer about availability. Obtain the NCD source sensor data from the Newer College Dataset authors.

Run the prepared-input pipeline from the repository root in a Linux environment using python -m research.multifloor.run --submaps <submap_directory> --vision-input <vision_directory> --runtime-config configs/runtime.yaml --camera-config configs/camera.yaml --output <output_directory> --stage all. Required inputs are poses.json, local LiDAR submaps, indexed four-camera images and masks, and global_dinov2.npz descriptors. The dependency record is research/requirements.txt.

For result verification, install NumPy and SciPy and run python tools/check_release.py and python tools/rescore_ncd.py. These commands check the release-file hashes, floor-height calculations, loop distribution, and NCD scores. The expected outputs are 17 single-storey loops; 26 four-storey loops distributed as 11, 3, 7, and 5 across F1–F4; a 3.83 cm mean absolute adjacent-floor height error; and NCD retrieval counts of 30/31 at rank 1 and 31/31 at rank 5. The complete expected-output checklist is provided in REPRODUCIBILITY.md.

Supplementary Note: Abbreviations

AP3P	algebraic perspective-three-point
BTC	binary and triangle combined descriptor
ESIKF	error-state iterated Kalman filter
FOV	field of view
GeM	generalized-mean pooling
GNC	graduated non-convexity
GTSAM	Georgia Tech Smoothing and Mapping
ICP	iterative closest point
IMU	inertial measurement unit
LiDAR	light detection and ranging

AP3P	algebraic perspective-three-point
LIVO	LiDAR–visual–inertial odometry
NCD	Newer College Dataset
NUC	Next Unit of Computing
PnP	perspective-n-point
RANSAC	random sample consensus
RMSE	root-mean-square error
RTAB-Map	Real-Time Appearance-Based Mapping
SC	Scan Context
SE(3)	special Euclidean group in three dimensions
SLAM	simultaneous localization and mapping
UAV	unmanned aerial vehicle
ViT	vision transformer
VPR	visual place recognition